# Proof

Questions 1–18

**2.44 — Question 1.** For each of the following statements, write the negation:

a) The cat is black.
b) Some cats are black.
c) Some cats are not black.
d) Line $A$ is parallel to line $B$.
e) The probability has increased.
f) $x \geq 0$
g) $x = 2$
h) $a$ is irrational.
i) $n$ is even and prime.

**2.44 — Question 2.** For each of the following statements, write the negation:

a) For all integers, $n$, either $n$ is positive or $n$ is negative.
b) $n$ is a number with at least 3 distinct prime factors.
c) Every prime number is odd.
d) The sum of a rational number and an irrational number is always irrational.
e) If $n^2$ is even, then $n$ is even.

**2.44 — Question 3.** A student is attempting to prove by contradiction that if $p$ and $q$ are positive integers, that $\frac{p}{q} + \frac{q}{p} \geq 2$
Their first line of working states:

> Given that $p$ and $q$ are negative numbers, $\frac{p}{q} + \frac{q}{p} \geq 2$

Correct their first line.

**2.44 — Question 4.** Prove that the following numbers are irrational:

a) $\sqrt{2}$
b) $\sqrt{3}$
c) $\sqrt{5}$

**2.44 — Question 5.** Prove that $\sqrt[3]{5}$ is an irrational number.

**2.44 — Question 6.** Prove that $1 + \sqrt{2}$ is irrational.

**2.44 — Question 7.** Prove that if $x^2 - 5x + 6 < 0$ then $2 < x < 3$

**2.44 — Question 8.** Prove that there is no greatest multiple of 3

**2.44 — Question 9.** Prove that there are infinitely many prime numbers.

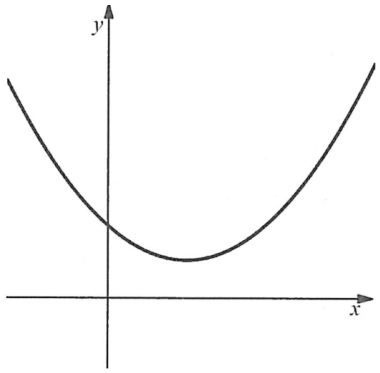
**2.44 — Question 10.** Prove that the difference between a rational number and an irrational number is always irrational.

**2.44 — Question 11.**

a) Prove that the product of a positive rational number and a positive irrational number is irrational

b) Hence, state the values for which it is not true that the product of a rational and an irrational number is irrational.

**2.44 — Question 12.** Lynn has sketched the graph of $f(x) = kx^2 - 3kx + 8$ which can be seen below.




She claims that the values of $k$ for which this is the correct sketch are $0 < k < \frac{32}{9}$
Use proof by contradiction to prove her claim.

**2.44 — Question 13.** Prove that for all $x > 0$, $x + \frac{1}{x} \geq 2$

**2.44 — Question 14.** Prove that for any real value of $a$ and $b$ then $a^2 + b^2 > 2ab$

**2.44 — Question 15.** Prove that $\log_{10} 3$ is an irrational number.

**2.44 — Question 16.**

a) Show that if $p^2 - 6q = 5$ then $p^2$ is odd

b) Hence, prove by contradiction that there are no integers $a$ and $b$ such that $a^2 - 6b = 5$

**2.44 — Question 17.** Prove that 3 is the only prime number which is 1 less than a square number.

**2.44 — Question 18. Prerequisite knowledge for Q18: Double Angle Formula** Prove that for all real $\theta$, $\cos\theta + \sin\theta \leq \sqrt{2}$